CS341 · COMPUTER GRAPHICS

Mock Midterm تدريب

20 درجة · نفس توزيع الاختبار

محارب

01 الأسئلة

Q1 TRUE / FALSE · 5 PTS

1. In a raster scan system, picture information is stored in the frame buffer. ( )

2. A plasma panel is a nonemissive display. ( )

3. Persistence is the time for the emitted light to decay to one-half of its original intensity. ( )

4. The difference between two points is a vector. ( )

5. In DDA, when |m| > 1 we step x by one unit and compute y. ( )

Q2 MCQ · 5 PTS

1. Which of the following is a hard-copy device?

a) Trackball b) Ink-jet printer c) Light pen d) Data glove

2. For crisp images, the dot pitch should be less than:

a) 0.13 mm b) 0.31 mm c) 3.1 mm d) 1.3 mm

3. Two vectors are perpendicular when their dot product is:

a) 1 b) −1 c) 0 d) Their length

4. The frequency at which a picture is redrawn on the screen is:

a) Resolution b) Persistence c) Refresh rate d) Dot pitch

5. In OpenGL, the function that sets the 2D world coordinate reference frame is:

a) glVertex2i b) gluOrtho2D c) glutInitWindowSize d) glClearColor

SHORT ANSWER · 4 PTS Q3

1. List four input devices.

2. Why do lines on a raster display look stair-stepped (jaggies)?

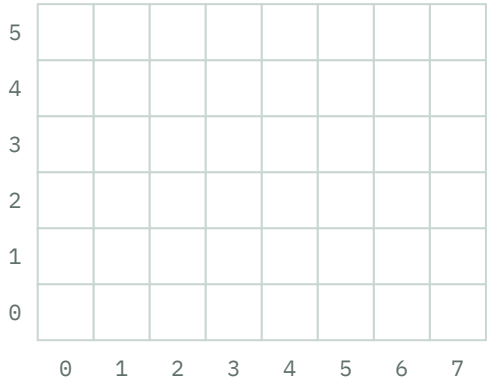
3. Compare random scan and raster scan (drawing, storage, resolution).

4. Compare active-matrix and passive-matrix displays.

DDA · 5 PTS **Q4**

Based on the DDA algorithm, calculate the points between the starting point (1,1) and the ending point (6,4). Fill the table, then draw the line using these points.

k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				



ASPECT RATIO · 1 PT

Q5

A screen is 8 inches wide and 6 inches high, with a resolution of 1024 × 768 points. Find the aspect ratio.